

Concurrent Management of an Exotic Species and Initial Restoration Efforts in Forests

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Abstract

One of the proximate results of forest fragmentation, and a cause of continued microenvironmental change and exacerbation of ecological problems, is increased invasions by weedy plant species. One such example is *Alliaria petiolata* (Brassicaceae), a serious pest threatening much of eastern North America. *Alliaria petiolata* impedes mitigation of fragmentation and restoration efforts because it tends to outcompete and possibly extirpate much of the native understory species on localized scales. As part of a strategy to address the problems of fragmented habitats, an experiment was conducted to determine whether *Sanguinaria canadensis* (Papaveraceae) could outcompete *A. petiolata*. Using an additive design, I transplanted *S. canadensis* at densities of 0, 1, 2, 3, 5, 7, 9, 11, 15, and 20 ramets/m² in 1997 and allowed them to interact with initial *A. petiolata* densities of 128 seedlings

and 31 rosettes/m². As of 2000, multivariate analyses of variance with repeated measures and simple analyses of variance indicated that initial *S. canadensis* densities of as little as 5 ramets/m² suppressed *A. petiolata*. Initial *S. canadensis* densities of 9 and 11 ramets/m² resulted in the lowest numbers of late-spring seedlings, numbers and sizes of year 1 and 2 rosettes, numbers and gross areas of stem leaves, numbers of flowering individuals, number of flowers, number of fruits (siliques), and height at flowering. While it remains to be tested whether this will continue and if the reestablishment of *S. canadensis* will help reassemble forest ecosystems, the experiments indicated that transplanting *S. canadensis* was effective at mitigating the spread of *A. petiolata*.

Key words: *Alliaria petiolata*, competition, edge effects, forest fragmentation, *Sanguinaria canadensis*.

Introduction

One impact of habitat fragmentation is increasing the relative length of the edges that surround an interior forest. The microenvironment in edges changes more rapidly than the historical ecotones surrounding the forest interior, and this creates new selection pressures that are unfavorable to many native species (e.g., Cadenasso et al. 1997). Exotic species may be adapted to edge microenvironments, and fragmentation facilitates invasion (e.g., Nuzzo 1991; Ehrenfeld & Toth 1997; Knapp & Canham 2000; Webb et al. 2001). The exotics tend to dominate, suppressing or even extirpating most native competitors, and alter the ecosystem structure and functions, e.g., gradually eliminate larger trees and shrubs by preventing recruitment, suppress or change mycorrhizae and other microflora and microfauna, and alter pH, cation exchange capacity, and nutrient cycling (Matlack 1993, 1994a, 1994b; Luken et al. 1997; Kourtev et al. 1998, 2002; Gould & Gorchoff 2000; Joly 2000; Ehrenfeld et al. 2001; Crooks 2002; Chornesky & Randall 2003). Ultimately, exotics may help make edge effects and fragmentation a self-organizing process (e.g., Murphy & Martin 2001; Collier et al. 2002).

Of particular concern is that exotic species may extirpate spring ephemerals comprising the "vernal dam." This term refers to species that prevent excess loss of nutrients in spring, following decomposition (Muller & Bormann 1976; Muller 1978; Blank et al. 1980; Zak et al. 1990; Eickmeier & Schussler 1993; Anderson & Eickmeier 1998, 2000). The loss of the spring ephemerals from fragmentation and concomitant invasion by competitive exotic species can mean that the nutrients are exploited by invasives and/or the nutrient cycle changes in a manner detrimental to the forest ecosystem (see Meter et al. 1995; Ehrenfeld et al. 2001; Honnay et al. 2002b; Booth et al. 2003, for general discussion).

A likely threat to spring ephemerals (and woodlands in general) is from the facultative biennial known as Garlic mustard (*Alliaria petiolata* (L.) Cavers & Grande, Brassicaceae). This species possesses phenotypic plasticity and an ability to adapt quickly. This makes it highly competitive (for light, nutrients, and perhaps, water), and it can eliminate most of the aboveground populations of most native species (the last extensive review was by Cavers et al. 1979). There is an extensive literature on *A. petiolata* (Nuzzo 1991, 1993a, 1993b, 1994, 1999; Anderson et al. 1996; Cruden et al. 1996; Nuzzo et al. 1996; McCarthy 1997; Beyers & Quinn 1998; McCarthy & Hanson 1998; Susko & Lovett Doust 1998, 1999, 2000; Dhillon & Anderson 1999; Drayton & Primack 1999; Meekins & McCarthy 1999, 2001, 2002; Luken & Shea 2000; Meekins et al. 2001; Roberts & Anderson 2001).

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When an exotic species like *A. petiolata* invades and dominates, it tends to alter the microclimate to its own benefit. Directly, *A. petiolata* changes the mycorrhizal community and perhaps the gross chemistry of soil via allelopathy (McCarthy & Hanson 1998; Meekins & McCarthy 1999, 2000). *Alliaria petiolata* outcompetes herbs, shrubs, and trees, mainly at juvenile stages (Meekins & McCarthy 1999), such that the forest edge becomes devoid of natives that used to regenerate a more complex and shaded structure. The open edges that *A. petiolata* colonize and then exacerbate leave the forests more vulnerable to windthrow and to more permanent changes in temperature, humidity, and light that promote further invasion of exotics and retreat of natives adapted to forest environments. Because *A. petiolata* began to dominate fragmented forests in the Region of Waterloo (Ontario, Canada), I recorded anecdotal observations of its impact and whether any native species seemed to be less vulnerable or somehow seemed to recover. By 1996, I noticed that one native species in particular seemed to fit this profile—Bloodroot (*Sanguinaria canadensis* L., Papaveraceae).

Sanguinaria canadensis is a perennial that has relatively large, fast-emerging shoots that rise to about 0.5 m at flowering and palmate leaves up to 30 cm in width. It reproduces asexually via rhizomes and sexually via autogamy and facultative xenogamy, with “solitary bee” (*Andrena* L.) species as an agent for pollination (Judd 1977). Seeds are dispersed, and helped to germinate, by ants (Schemske 1978; Pudlo et al. 1980; Beattie 1982; Lyon 1992; Lobstein & Rockwood 1993; Liu & Heithaus 1994; Marino et al. 1997; Bono & Heithaus 2002). There is some support for my anecdotal observation that *S. canadensis* seems to be one of the few species that can withstand invasion by *A. petiolata*. For *S. canadensis*, there is increased asexual reproduction and leaves get larger with exposure to increased light and nutrients typical of fragmented forests and their edges (Marino et al. 1997). Although seed production and dispersal is not as plastic and does not increase in fragmented areas, it still produces up to 500 viable seeds/ramet that are added to the seed bank (Marino et al. 1997). *Sanguinaria canadensis* also has alkaloids that help defend against herbivores and possibly other plants (Bennett et al. 1990; Salmore & Hunter 2001a, 2001b).

Although the long-term challenge in solving the impacts of exotic species is to halt fragmentation on local and landscape scales, the immediate challenge is that remnant forests must be protected and restored before too many species are extirpated. One approach is to simultaneously manage exotic invaders and restore species; care must be taken because exotic removal is not always the best course of action in general (Zavaleta et al. 2001; D’Antonio & Meyerson 2002), and it may just create opportunities for other exotic species (Webb et al. 2001). It may be useful to reintroduce native species that compete successfully with exotics as a useful, albeit small, first step to restoring forest ecosystems. In this article, I discuss the results of an experiment that tested whether the establishment of

S. canadensis could be used to successfully compete with exotic *A. petiolata*.

Methods

Study Site

The site chosen was a degraded woodlot (total area = 31,916 m²) located on a farm approximately 16 km northwest of Waterloo, Ontario, Canada. Experiments were conducted on the first 50 m the woodlot near the identifiable edge of the northern face. In this part of Ontario, prevailing north and west winds from Lake Huron (approximately 100 km away) tend to be the most damaging and leave these faces of woodlots most vulnerable. The choice of experimental locations within the site was based on the criteria that they had large populations of *Alliaria petiolata* but few or no individuals of *Sanguinaria canadensis*. This may have been because *S. canadensis* was competitively excluded, the microenvironment could not support this species, or it may have been a stochastic event. There were few other species present at any time of the year in the plots; hence, there was no need to remove them. The collective background density was a maximum of 4 plants/m² of some or all of Common spring beauty (*Claytonia virginica* L.), Yellow trout lily (*Erythronium americanum* Ker.), Dame’s rocket (*Hesperis matronalis* L.), Canada avens (*Geum canadense* L.), Herb-robert (*Geranium robertianum* L.), Canada bluegrass or wiregrass (*Poa compressa* L.), and Woodland speargrass (*Poa nemoralis* L.). There were no more than 1 shoot/m² of extant seedlings of Sugar maple (*Acer saccharum* L.), Common beech (*Fagus grandifolia* Ehrh.), and White oak (*Quercus alba* L.). These were left intact. There were no other shrubs, saplings, or mature trees rooted in any of the plots.

Experimental Design

An additive series design was used to test the effects of competition from *S. canadensis* on both seedlings and rosettes of *A. petiolata*. An additive series was chosen because the experiment’s purpose was to test the response of only one species (*A. petiolata*) to a competitor (*S. canadensis*) within the context of restoration ecology. This form of additive design keeps intraspecific competition within *A. petiolata* constant and eases measurement of interspecific competition between *A. petiolata* and *S. canadensis*. However, it does not allow quantitative measurement of intraspecific competition within *S. canadensis*. Because it is more rigorous, a response surface design (Goldberg & Scheiner 1993; Inouye 2001) was considered but rejected because of the very reasons these authors warn about: insufficient personnel, time, and logistical support to plant so many density combinations.

The experiments were based on a randomized complete block design of four replicates, with each experimental unit surveyed as a square measuring 1 × 1 m, with 2-m

guard rows surrounding each unit. Arbitrarily chosen rhizomes of *S. canadensis* growing near the experimental plots (at a distance of 25–75 m) were tagged in spring and summer 1997. In autumn of that year, the rhizomes were harvested inside a clod of soil, removed briefly to measure length and biomass in a laboratory, and quickly transplanted within the same clod of the soil in which they originated. This allowed assessment of initial size, facilitated survival, and reduced transplant shock. The transplants were assessed in April 1998 to ensure that none failed to leaf; in fact, all leafed in spring and survived the transplant the previous autumn.

Ten densities of rhizomes, representing ramets, of the putative competitor *S. canadensis* were transplanted: 0 (control), 1, 2, 3, 5, 7, 9, 11, 15, and 20 per m². The mean density of *S. canadensis* at the experimental site and at five others sampled within a 30-km radius was 3.7 (± 1.8) per m², with the highest mean density recorded as 8.2 (± 1.6) per m² ($N = 25$ randomly selected 1-m² quadrats at each of the six sites). Because competitive effects often become nonlinear at extremely low densities (e.g., Goldberg & Scheiner 1993), more emphasis was placed on densities below this mean and standard error. The two highest densities chosen were beyond what is found locally, but these densities might be useful or even necessary in restoration approaches designed to manage *A. petiolata*. The pattern of planting was kept consistent across the replicates. The 1 ramet/m² density had the rhizome transplanted to the middle of the plot. Subsequent densities kept distances both between the transplanted rhizomes and between transplants and the edges of each replicate plot as equidistant as possible within the confines of the geometric shape, a square, that I used.

The initial (1997) density of seedlings of *A. petiolata* (the focal species) was kept constant at 128 per m² because this was the mean value observed in nearby sites within Waterloo Region in another study (Murphy, unpublished data) and is not atypical of other infested areas in North America—Meekins and McCarthy (2000). For similar reasons, the number of rosettes of *A. petiolata* was kept constant at 31 per m². The initial rosette sizes of *A. petiolata* could not be standardized in a field study but were all within the standard deviation (2.4 cm) of the mean rosette size (13.2 cm) found locally in 1996–1997.

The population densities of *A. petiolata* at the experimental sites were high enough that no additional individuals needed to be transplanted into the plots; hence, the initial densities were achieved by removing excess seedlings or rosettes before the experiment's initial surveys on 1 April 1997. It was not possible to standardize the spatial location of the remaining plants, though attempts were made to select forest sites with a relatively even coverage of *A. petiolata*; hence, any replicate would have similar spatial configurations of plants—the spatial effects are currently being tested in a longer-term study. All the remaining individuals of *A. petiolata* were identified with an aluminum plant marker.

Variables Measured

In total, I measured 20 demographic and phenological responses. To offer some indication of how *S. canadensis* responded over the 4 years reported here, I included measurements of the number of leaves of *S. canadensis* emerging successfully. Leafing phenologies of *S. canadensis* and *A. petiolata* were recorded for each replicate. Because the study's additive design meant it was focused mainly on competitive response of *A. petiolata*, many characters were measured:

- The numbers of seedlings, number of rosettes, and size of rosettes counted in spring 1997–2000 began from the first emerging seedling and visible rosette—locally, this is usually late March or early April when the snow melts enough for the new seedlings of *A. petiolata* to emerge and year 2 rosettes to become visible. This continued through the first-leaf emergence of *S. canadensis*—typically, this was near the third week of April. I called this the “early-spring” observation.
- The same observations were made at the end of the leafing period of *S. canadensis* in 1997–2000. I called this the “late-spring” observation. Locally, it typically lasted for 5 days during early to mid-May. I monitored populations throughout the growing season, so the decision as to when this leafing period ended actually was retroactive and more accurate that way. By mid-May, most trees have not yet leafed out in Waterloo Region, so most shading is related to understory species.
- The same observations were made during the last week of September in 1997–2000; I refer to this as the “autumn” observation. The rosettes measured here were not the same individuals as measured in spring; these represented the surviving year 1 seedlings and those that survived over winter would be counted again in spring as year 2 rosettes.
- The cumulative number of bolted and blooming individuals, as based on a weekly survey. All individuals that bolted also bloomed, so these numbers were the same.
- On individuals, if they bolted and bloomed: number of leaves and gross leaf area on stem leaves (calculated as 0.5 base $\times$ height) when first flowering was observed, cumulative number of racemes, flowers and siliques, and height when flowering concluded.

Neither biomass of tissues nor viable seed set in either species was measured. This was because the destructive sampling and soil disturbance required would have affected the population dynamics of both the species and made it difficult to assess the impact of planting *S. canadensis* on *A. petiolata*. Because this experiment will continue to run indefinitely, no purposeful disturbance or destruction was used.

The rate of change in the numbers of seedlings and rosettes in *A. petiolata* was calculated for the periods between the first week of April and end of leafing in *S. canadensis*, end of leafing in *S. canadensis* and the last week of September, and the end of September and first

week of April (these were mainly the overwintering year 1 cohort survivors, but a few new seedlings or rosettes emerged or developed during this time period). The rate of change/production in the cumulative number of bolting and blooming individuals of *A. petiolata* was calculated for the entire April–September observation period.

Analyses

On the advice of reviewers of an earlier draft of this article, with corroboration from von Ende (1993), I restricted the number of dependent variables analyzed and presented here. Although I measured 20 variables in *A. petiolata*, the power of analysis would be diminished if all the variables were included in the preferred analysis—a multivariate repeated measures analysis of variance (MANOVA) to compare the effects of different density treatments (Gurevitch & Chester 1986; Scheiner 1993; von Ende 1993). As a result, I ran constrained MANOVAs on smaller numbers of variables and separated variables on the basis of whether they reflected population densities versus the individual characteristics like flower number and also constrained the analysis by grouping them based on the time of year a variable was measured. For the final analyses presented in Results, I ran MANOVAs on already identified significant variables and for responses of population densities and then for individual characteristics. I did not separate them by time of year measured in this part of the analysis. These excluded variables still were analyzed in a second comparison (see last paragraph in this section).

Relevant to the MANOVA model, densities were manipulated only once; the measures were repeated over time on a per-plot basis, and interactions were expected to occur on an individual basis. However, the numbers of seedlings, rosettes, and flowering individuals were expected to be nonindependent as well. For all the repeated measures, there was one within-subjects factor (time, measured in weeks since the experiment was initiated) and one between-subjects factor (density of *S. canadensis* when the experiment was initiated). The within-subjects factor of weeks considered time “0” to be the first week of observation in April 1997 and counted all weeks thereafter, though there were no observations between 1 October and 1 April.

Once the MANOVA tests were completed, I ran a series of one-way analyses of variance (ANOVAs), with Tukey’s test for honestly significant differences ($HSD_{0.05}$) on the 2000 data to determine whether the variables measured on *A. petiolata* differed significantly, based on the initial density in 1997 of *S. canadensis*. These were completed for all variables because they are independent ANOVAs and will not be affected by inclusion of too many dependent variables.

Results

Although seedling emergence of *Alliaria petiolata* was relatively early, the leafing phenologies of year 2 *A. petiolata*

and ramets of *Sanguinaria canadensis* overlapped (Fig. 1). With time the numbers of leaves emerging from *S. canadensis* rose most rapidly from the ramets planted in initial densities of 7, 9, and 11 ramets/m² (Fig. 2).

The results for the competition experiments generally showed that the initial densities of *S. canadensis* had different effects on *A. petiolata* in subsequent years (Table 1). The significant time $\times$ density interactions indicated that the effects of initial densities changed over time as the populations of *A. petiolata* were affected by competition with *S. canadensis* after the first season. The only variables not affected were number of seedlings in early spring, number of seedlings in autumn, change in numbers of seedlings from autumn to spring, change in number of year 2 rosettes from early to late spring. These were analyzed only in the unreported initial analyses and are excluded from Table 1. Generally, initial *S. canadensis* densities of 7, 9, and 11 ramets/m² suppressed *A. petiolata* most (Tables 1–3; Figs. 3 & 4). When within-year ANOVAs were run for the 2000 data, there were no significant differences on most variables measured for *A. petiolata* among initial *S. canadensis* densities of 9, 11, 15, and 20 ramets/m², though 9 and 11 ramets/m² tended to be the most effective, e.g., see the response of the numbers of seedlings in late spring and leaf area in Tables 1–3.

At lower initial densities (3 ramets/m² or less), *S. canadensis* usually could not compete with *A. petiolata*. At the two highest initial densities (15 and 20 ramets/m²), *S. canadensis* ultimately declined, probably through density-dependent intraspecific competition, and more seedlings and rosettes of *A. petiolata* survived to compete with *S. canadensis*. By 2000, this meant that the higher densities of *S. canadensis* provided no additional benefit in suppressing *A. petiolata* and fared slightly, but not significantly, worse than densities of 5, 7, 9, and 11 ramets/m².

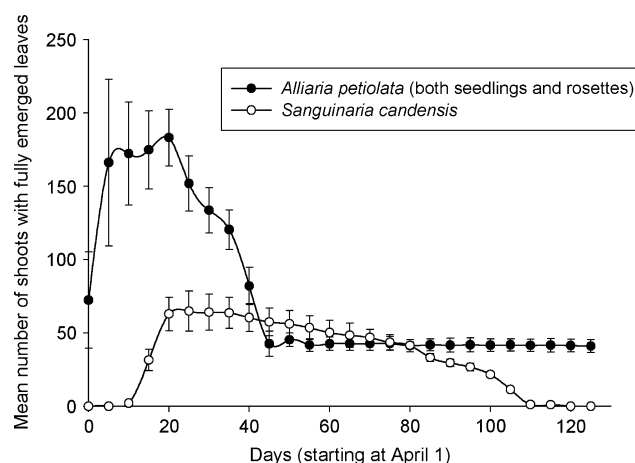


Figure 1. Leafing phenologies for *Alliaria petiolata* and *Sanguinaria canadensis* during the 125 days following 1 April inclusive. Data are reported as the means and standard errors across all years (1997–2000) in the number of first and second year genets (*A. petiolata*) bearing fully emerged initial or basal rosette leaves and the number of ramets (*S. canadensis*) bearing fully emerged basal leaves.

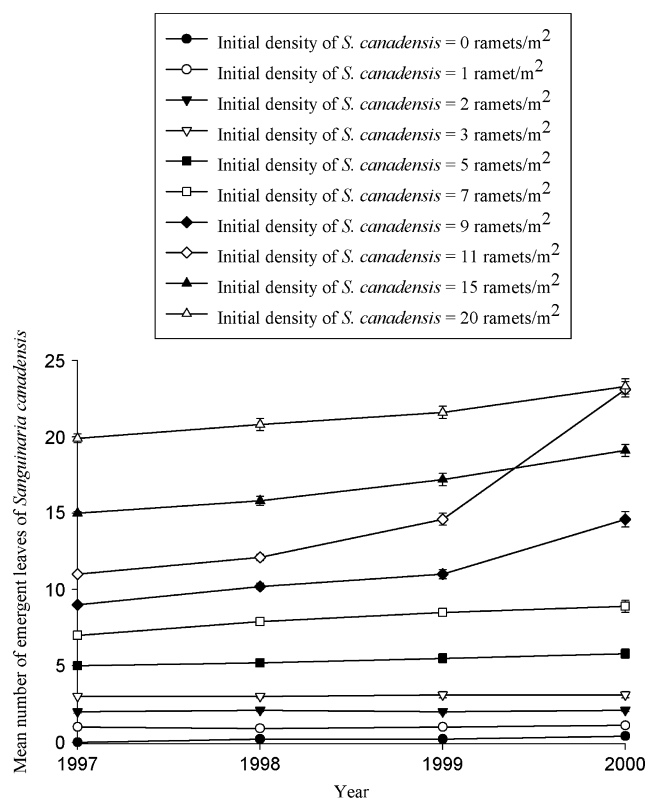


Figure 2. Mean number of emergent leaves from ramets of *Sanguinaria canadensis* from 1997 to 2000.

The increase in the number of leaves of *S. canadensis* at initial intermediate densities of 9 and 11 ramets/m² (Fig. 2) to levels equal to or near the highest initial density of 20 ramets/m² indicates that these intermediate densities were most useful in competing with *A. petiolata*. However, I do not suggest that this one measure is a substitute for a complete response surface test for intra- and inter-specific competition affecting *S. canadensis*.

Early-spring emergence of seedlings in *A. petiolata* remained relatively constant, probably indicating that an existing, extensive seed bank persisted throughout and was not affected after only 4 years. This was not surprising

because soil cores of seven woodlots of similar area and within 5 km of my study site have revealed that the mean number of seeds of *A. petiolata* was 2,567 (± 478) per m³. Additionally, *S. canadensis* probably leafs too late to suppress most seed germination and seedling emergence in *A. petiolata*. By the time the first leaves of *S. canadensis* appeared, most of the seedlings of *A. petiolata* have emerged and leafed (Fig. 1). However, leafing at (initial) densities of *S. canadensis* above 7 ramets/m² was early enough to kill significantly more seedlings of *A. petiolata* before they matured; both the numbers of late-spring seedlings and the change in numbers of seedlings between early and late spring declined significantly at these densities by 2000 (Tables 1 & 3; Fig. 3). By summer, however, the leaves of *S. canadensis* senesced and, with such a large seed bank, there were no differences between density treatments in the numbers of autumn seedlings that emerged during summer or change in numbers of seedlings of *A. petiolata* between late spring and autumn (Tables 1 & 3).

The numbers and sizes of rosettes of *A. petiolata* were affected more seriously in the sense that, by 2000, the initial densities of even 5 ramets/m² of *S. canadensis* significantly harmed early-spring, late-spring, and autumn rosettes (Tables 2 & 3; Figs. 3 & 4). Below this density, rosette numbers and sizes increased over time (Tables 2 & 3; Figs. 3 & 4).

Because there were no significant changes in the numbers of rosettes between year 1 autumn and year 2 early spring or year 2 early spring and year 2 late spring, it means that the impact was probably related to the effect of *S. canadensis* on seedlings as opposed to any direct suppression of rosettes of *A. petiolata*. Higher densities of *S. canadensis* likely killed or harmed enough seedlings of *A. petiolata* so that there were fewer survivors to produce year 1 rosettes, and those that survived produced smaller rosettes (Tables 1 & 2; Figs. 3 & 4). The autumn data included later emerging *A. petiolata* that escaped competition from *S. canadensis*, but any later emergents were not great in numbers. Although I have not reported these in detail, a maximum of 4.2% year 1 rosettes and seedlings came from individuals that emerged after mid-June in any year.

Table 1. MANOVA testing responses of population characteristics of *Alliaria petiolata* to 10 different densities of *Sanguinaria canadensis*.

Variable	Density			Time			Time $\times$ Density		
	MS	F	p	Pillai	F	p	Pillai	F	p
No. of seedlings in late spring	43.28	22.79	***	0.74	21.25	***	0.79	26.53	***
Change in no. of seedlings early to late spring	40.17	29.25	***	0.69	17.10	***	0.75	24.04	***
No. of year 1 rosettes in autumn	29.11	24.80	***	0.81	26.92	***	0.76	22.38	***
No. of year 2 rosettes in early spring	20.03	13.42	***	0.76	22.46	***	0.70	19.79	***
No. of year 2 rosettes in late spring	19.27	12.81	***	0.62	15.54	***	0.69	18.46	***

Tests were based on data collected from 40 plots observed during the first week of April through the last week of September from 1997 to 2000. Results are reported only for the combined MANOVA using variables reported as significant in constrained MANOVAs (see text). For density of *S. canadensis* (the between-subjects effect), data are reported as the standard used for repeated-measured ANOVA (MS, F, p). For time and time $\times$ density (within-subjects effects), data are reported using F, p, and Pillai's Trace because the latter is the actual test for significant differences of the repeated factor of time (and time $\times$ density) within subjects. MS=mean square error.

Table 2. MANOVA testing responses of individual characteristics of *Alliaria petiolata* to 10 different densities of *Sanguinaria canadensis*.

Variable	Density			Time			Time $\times$ Density		
	MS	F	p	Pillai	F	p	Pillai	F	p
No. of stem leaves	46.14	30.11	***	0.68	49.25	***	0.60	17.87	***
Leaf area	37.49	27.88	***	0.88	79.17	***	0.64	15.09	***
Cumulative no. of flowering individuals	33.60	30.16	***	0.89	27.10	***	0.91	46.55	***
No. of flowers	39.32	29.45	***	0.84	76.24	***	0.52	11.18	***
No. of siliques	52.07	47.71	***	0.90	80.94	***	0.74	18.43	***
Height at flowering	28.89	25.12	***	0.77	64.07	***	0.54	10.99	***

Tests were based on data collected from 40 plots observed during the first week of April through the last week of September from 1997 to 2000. Results are reported only for the combined MANOVA using variables reported as significant in constrained MANOVAs (see text). For density of *S. canadensis* (the between-subjects effect), data are reported as the standard used for repeated-measured ANOVA (MS, *F*, *p*). For time and time $\times$ density (within-subjects effects), data are reported using *F*, *p*, and Pillai's Trace because the latter is the actual test for significant differences of the repeated factor of time (and time $\times$ density) within subjects. MS=mean square error.

The number of stem leaves, gross leaf area, number of flowers, number of siliques, and height at final flowering were all significantly suppressed by *S. canadensis* at intermediate and higher densities of 7, 9, 11, 15, and 20 ramets/m² (Tables 1 & 3). These are all good indicators that density-dependent competition from *S. canadensis* had a significant effect on *A. petiolata*. It appears that intermediate and higher densities of *S. canadensis* deprived *A. petiolata* of sufficient resources to produce large leaves, flowers, and fruits, indicating that both vegetative and reproductive allocations were affected. The inability of *A. petiolata* to allocate more resources to grow taller under intermediate and higher densities of *S. canadensis* means that it either had insufficient resources or could not reallocate resources quickly enough when in competition.

Discussion

For ecological restoration, the results suggest that planting intermediate densities of *Sanguinaria canadensis* is a good strategy in managing invasions by *Alliaria petiolata*. Although the lack of an effect on early- and late-season numbers of seedlings might seem to indicate that the situation is hopeless, this is true only if the goal is to quickly reduce short-term populations of *A. petiolata*. This was not the purpose here. Even though the spring populations of *A. petiolata* remained high, they did not worsen when densities of five or more ramets per square meter of *S. canadensis* were transplanted. Given the reduced fruit set, a long-term implication is that populations of *A. petiolata* will decrease locally as long as immigration of new seeds does not counter this. In a small woodlot, this is likely to occur with restoration efforts. In larger, contiguous woodland infested with *A. petiolata*, this would be more difficult (e.g., Drayton & Primack 1999; Nuzzo 1999).

All else being equal, densities of 9 and 11 ramets/m² of *S. canadensis* resulted in more successful competition with *A. petiolata*. But for what resources was there competition? I did not design the experiment to test directly for the mechanism of competition, but the literature offers strong ideas. Although competition can be for any limited

resource, most understory plants tend to compete for light because nutrients and water are normally plentiful (Lapointe 2001; Rothstein & Zak 2001; Tremblay & Larocque 2001; Hull 2002; Routhier & Lapointe 2002). Previous evidence for competitive superiority of *A. petiolata* indicates that its early emergence and competition for light are most important (Meekins & McCarthy 1999, 2000). Ultimately, a more complicated competition experiment will be useful in elucidating mechanisms, that is, test for the complexities of different resources, environmental heterogeneity, size- and density-dependent competition, genotype by environment interactions. However, the scope of my experiment already was labor intensive and complex; hence, this hypothetical experiment may be too unwieldy.

My study has determined that some densities of *S. canadensis* were indeed successful at halting the dominance of *A. petiolata* in early to late spring, and *S. canadensis* established itself to begin ecological restoration. This success may provide opportunity for other native species to reestablish from nearby propagules sources and increase the impact of ecological restoration, but thus remains to be tested explicitly. In my study, there is no evidence yet of regeneration of other native species from the existing propagules. By 2000, I had not detected any increase in other native species or other exotics in any treatments; the data were not reported because of the lack of any large numbers of individuals and the subsequent lack of any trends. The absence of any significant reestablishment of other natives may be explained by their inability to compete with the still high densities of *A. petiolata*, the generally higher densities of *S. canadensis*, a lack of seed bank or immigration sources, and an ecosystem that cannot support their natural regeneration, especially in terms of the mycorrhizal and decomposition communities (Baar & Kuyper 1998; Emmerling 1998; Ehrlén & Eriksson 2000; Butaye et al. 2002; Tyler et al. 2002). Of course, it may be a simple case if being too early in ecological restoration to expect and detect any changes for species other than the transplanted *S. canadensis*.

Pragmatically, transplanting densities of *S. canadensis* at 9 or 11 ramets/m² generally would be cost and labor

Table 3. ANOVAs testing for differences in response of *Alliaria petiolata* in 2000 to the 10 different initial (1997) *Sanguinaria canadensis* density treatments.

2000 Response of <i>A. petiolata</i>	Initial Density in 1997 of <i>S. canadensis</i> (Ramets/m ²)									
	0	1	2	3	5	7	9	11	15	20
No. of seedlings in early spring	142.9 (32.7)a	127.8 (36.6)a	118.1 (41.2)a	137.1 (29.6)a	118.6 (33.4)a	113.8 (31.2)a	129.3 (45.9)a	122.7 (43.7)a	121.3 (35.1)a	117.9 (38.0)a
No. of seedlings in late spring	95.9 (14.8)a	76.9 (16.2)ab	65.7 (15.9)b	59.2 (14.1)bc	41.2 (13.7)cd	34.1 (15.9)de	9.2 (5.7)f	8.7 (3.2)f	23.5 (8.8)e	27.1 (7.2)e
No. of seedlings in autumn	10.5 (6.2)a	10.5 (7.8)a	10.1 (5.9)a	11.7 (9.2)a	7.5 (6.7)a	10.7 (7.4)a	12.4 (8.9)a	12.8 (10.2)a	12.5 (7.5)a	12.2 (9.1)a
No. of year 1 rosettes in autumn	53.6 (18.4)a	50.2 (15.7)a	41.2 (13.3)ab	36.2 (11.9)bc	24.3 (9.4)cd	21.6 (8.2)cd	15.1 (6.2)d	13.6 (5.7)d	27.6 (8.4)cd	26.7 (9.3)cd
No. of year 2 rosettes in early spring	46.8 (15.5)a	47.1 (17.3)a	37.2 (9.2)ab	35.9 (9.3)ab	26.7 (9.9)bc	22.6 (7.4)bc	16.9 (6.5)c	14.5 (5.3)c	28.1 (7.2)bc	26.2 (7.4)bc
No. of year 2 rosettes in late spring	46.2 (10.8)a	45.3 (11.8)a	35.4 (8.7)ab	34.2 (8.5)ab	25.1 (8.0)bc	20.8 (8.6)bc	16.4 (4.3)c	13.2 (5.0)c	27.0 (6.8)bc	24.7 (7.2)bc
Year 1 rosette size in autumn	31.2 (6.7)a	29.8 (7.2)a	27.9 (6.5)ab	20.6 (4.9)b	10.2 (3.7)c	5.8 (1.2)d	5.1 (1.6)d	4.7 (1.4)d	8.1 (2.3)cd	7.8 (2.6)cd
Year 2 rosette size in early spring	31.4 (6.4)a	29.9 (7.1)a	28.2 (6.3)ab	21.5 (4.7)b	10.3 (3.3)c	5.6 (1.0)d	5.4 (1.7)d	4.6 (1.4)d	7.9 (2.0)cd	7.7 (2.7)cd
Year 2 rosette size in late spring	31.4 (6.2)a	32.5 (7.4)a	29.1 (6.5)ab	27.0 (4.9)b	11.5 (3.5)c	5.6 (1.1)d	5.3 (1.7)d	4.6 (1.5)d	8.0 (2.1)cd	8.2 (2.8)cd
No. of stem leaves	11.9 (3.2)a	11.7 (2.8)a	11.1 (2.9)a	9.7 (2.4)ab	7.9 (1.6)b	6.2 (1.4)bc	4.5 (1.4)c	4.1 (1.2)c	4.3 (1.5)c	4.6 (1.7)c
Leaf area	24.3 (6.1)a	22.6 (6.5)a	19.2 (5.4)a	17.2 (4.3)ab	12.1 (3.5)b	9.4 (2.9)b	8.8 (3.1)bc	5.9 (1.4)c	8.3 (1.9)bc	8.6 (2.2)bc
Cumulative no. of flowering individuals	41.3 (11.4)a	40.7 (10.3)a	32.3 (9.1)ab	33.6 (8.2)ab	22.7 (6.6)b	18.6 (5.3)bc	13.8 (4.4)c	13.2 (4.2)c	20.9 (5.5)bc	19.5 (5.1)bc
No. of racemes	2.6 (1.3)a	2.7 (1.2)a	2.2 (1.6)a	2.5 (1.0)a	2.1 (1.4)a	2.6 (1.2)a	2.8 (0.9)a	1.9 (1.3)a	2.2 (1.4)a	2.5 (1.2)a
No. of flowers	104.6 (31.7)a	105.2 (28.7)a	97.1 (29.2)a	80.2 (22.8)ab	67.3 (20.4)bc	46.3 (16.7)bc	41.5 (11.2)c	39.6 (12.1)c	47.4 (14.2)bc	50.2 (12.8)bc
No. of siliques	11.7 (2.7)a	10.4 (2.4)a	9.4 (2.3)a	9.0 (2.4)a	8.0 (2.1)ab	3.6 (1.8)c	2.5 (1.6)c	2.2 (1.7)c	6.2 (1.7)bc	7.0 (1.5)bc
Height at flowering	94.7 (31.2)a	92.4 (26.7)a	78.3 (24.3)ab	70.1 (19.7)ab	61.5 (16.8)bc	36.9 (14.3)bc	23.1 (10.1)c	20.5 (10.2)c	29.6 (12.2)c	30.1 (10.9)c

Tests were based on data collected from 40 plots observed during the first week of April through the last week of September from 1997 to 2000. Within each row, data followed by the same letter are not significantly different according to Tukey's test for HSD_{0.05}.

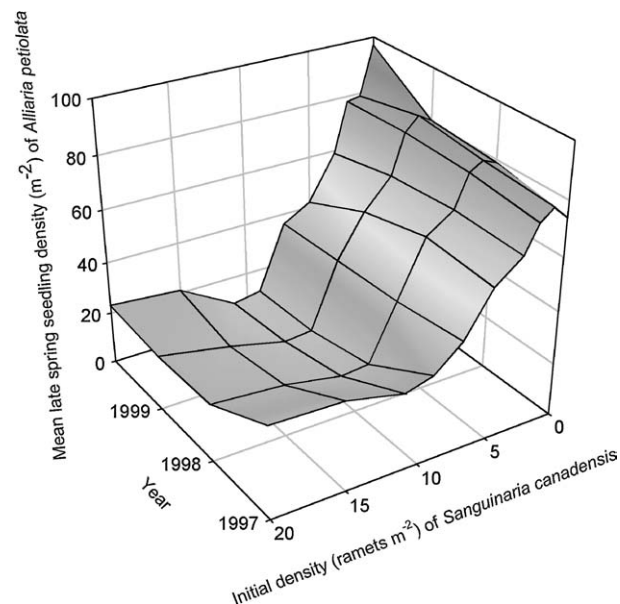


Figure 3. A comparison of mean late-spring seedling density (m⁻²) of *Alliaria petiolata* and initial density of ramets (m⁻²) of *Sanguinaria canadensis* during 1997–2000.

effective on a local scale of a couple of hundred meters at most. Ramets of *S. canadensis* may be rescued from nearby development, grown locally from a ramet or seed stock, or purchased locally for approximately US \$2.00 per 8-cm pot. The transplanting in my study required about 8 hours for 40 m² of plots, but this was because I

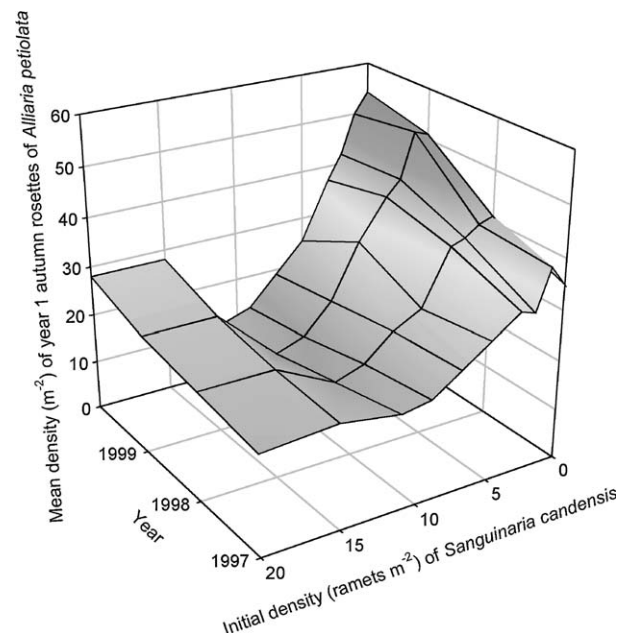


Figure 4. A comparison of mean year 1 rosette density (m⁻²) of *Alliaria petiolata* and initial density of ramets (m⁻²) of *Sanguinaria canadensis* during 1997–2000.

positioned the ramets in a consistent geometric pattern; it would take much less time with a team of volunteers who need only approximate spacing and densities to reduce intraspecific competition.

This is a proximate and local-scale solution that addresses immediate threats of exotic species and a lack of native species for ecological restoration. Longer-term, if many locales adopt the planting of species like *S. canadensis*, it may help establish local barriers that mitigate the spread of *A. petiolata*. Ultimately, there may be opportunity to test whether this mitigation occurs and whether there is creation of a possible nurse layer of an understory species. If so, a forest edge ecosystem may redevelop—one that is less hostile to native species, including the shrubs and trees that will give it a more impermeable structure.

If *S. canadensis* ultimately can be demonstrated to encourage regeneration of other native species and create nodes producing emigrants able to colonize other areas, then restoration, like invasion, may self-organize and become passive in that sense. One question is whether this self-organization actually will occur (Anand & Desrochers 2004). A second question is whether *S. canadensis* or any returning native species will begin to experience density-dependent herbivory or long-term problems related to fragmentation impacts on ant dispersal of its seeds (Ruhren & Handel 2000, 2003; Mitchell et al. 2002). Though it remains to be determined whether the necessary ecosystem functions will return and how much active restoration will be needed (Keddy & Drummond 1996; Ehrenfeld & Toth 1997; Luken et al. 1997; McLachlan & Bazely 2001; Honnay et al. 2002a), if there are simultaneous larger-scale efforts to conserve interior forest ecosystems and reduce fragmentation, then the rate of active and passive restoration of forest edges eventually will be able to catch up and overtake the rate of damage.

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